

MONGO DB

MongoDB Data types, Collections, Databases, and Documents.

Data Types:

- **Date:** Stores dates and times as a 64-bit integer representing milliseconds since the Unix epoch (January 1, 1970, 00:00:00 UTC). It's useful for timestamps, creation dates, etc.

Example (inserting a document with a date field):

```
db> db.products.insertOne({
...   name: "T-Shirt",
...   created_at: new Date() // Inserts the current date and time
... });
{
  acknowledged: true,
  insertedId: ObjectId('6669c1ee8aefb52a76cdcdf6')
}
db> show collections
candid
locations
products
student
student1
students
students_permission
db> |
```

- **Int32:** Represents 32-bit signed integers, ranging from -2,147,483,648 to 2,147,483,647. Use this for whole numbers within this range.

Example (storing product quantity):

```
db> db.products.insertOne({
...   name: "Shirt",
...   quantity: 10
... });
{
  acknowledged: true,
  insertedId: ObjectId('6669c22b8aefb52a76cdcdf7')
}
db> |
```

- **Decimal128:** Stores high-precision decimal numbers using 128 bits. It's ideal for financial calculations, currency values, etc., where exact precision is crucial.

Example (storing product price with decimals):

```
db> db.products.insertOne({
...   name: "Laptop",
...   price: Decimal128.fromString("1299.99")
... });
{
  acknowledged: true,
  insertedId: ObjectId('6669c2508aefb52a76cdcdf8')
}
db> |
```

- **Timestamp:** A 64-bit value consisting of two parts:
 - Seconds since the Unix epoch (similar to Date).
 - An optional incrementing ordinal for operations within a given second (less commonly used).

Note: While Timestamp exists, it's generally recommended to use Date for most date and time operations in MongoDB.

Collections, Databases, and Documents:

- **Database:** The top-level container in MongoDB that holds collections. You can create multiple databases within a single MongoDB instance.
- **Collection:** A group of related documents that share a similar schema (structure). Think of it as a table in a relational database. You can have multiple collections within a database.
- **Document:** A flexible data structure that stores information in key-value pairs. Each document can have different fields with different data types, allowing for rich and diverse data representation. Documents within a collection can vary in structure to some extent.

Example:

Consider a database named "eCommerce" with a collection named "products." Each document in this collection could represent a product:

```
{
  "_id": ObjectId("639a2b3c4d5e6f7890123456"),
  // Unique identifier for the document
  "name": "T-Shirt",
  "price": Decimal128.fromString("19.99"),
  "quantity": 50,
  "created_at": ISODate("2024-06-12T00:00:00Z")

  // Example using ISODate for a specific date
}
```

In this example:

- `_id` is a special field automatically generated by MongoDB to uniquely identify each document.
- Other fields (name, price, quantity, and `created_at`) have different data types to store specific product information.

KEY POINTS:

- Data types determine how data is stored and manipulated. Choose appropriate types based on your data.
- Collections logically group related documents.
- Databases organize collections for better structure.
- Documents hold your actual data, each with its own structure.

Install MongoDB Shell on Windows

MongoDB shell

Step 1:

Go to MongoDB download Page and click download as shown in the screenshot. It will be downloaded in your system. Double click on the file to run the installer.

MongoDB Shell Download

MongoDB Shell is the quickest way to connect to (and work with) MongoDB. Easily query data, configure settings, and execute other actions with this modern, extensible command-line interface – replete with syntax highlighting, intelligent autocomplete, contextual help, and error messages.

Compatibility Note: Red Hat Enterprise Linux (RHEL) 7, Amazon Linux 2, SUSE Linux Enterprise Server (SLES) 12, and Ubuntu 18.04 support is deprecated and might be removed in a later mongosh release.

Note: MongoDB Shell is an open source (Apache 2.0), standalone product developed separately from the MongoDB Server.

Step 2: Click Download

Version

2.2.6

▼

Platform

Windows x64 (10+)

▼

Package

zip

▼

Download

Copy link

More Options

Step 3: Click Next when the MongoDB Shell installation windows pops up. Extract all the files from mongosh folder. Next select the bin. When you select mongosh the shell window is opened.

mongosh-2.2.6-win32-x64					
File folder					
bin	File folder				16-05-2024 18:39
.sbom	JSON Source File	30 KB	No	350 KB 92%	16-05-2024 18:27
LICENSE-crypt-library	File	6 KB	No	18 KB 69%	16-05-2024 18:32
LICENSE-mongosh	File	4 KB	No	11 KB 67%	16-05-2024 18:32
mongosh.1.gz	WinZip	7 KB	No	7 KB 0%	16-05-2024 18:39
README	File	1 KB	No	1 KB 39%	16-05-2024 18:32
THIRD_PARTY_NOTICES	File	128 KB	No	1,862 KB 94%	16-05-2024 18:32

Step 4: Click on the mongosh , mongosh window is opened.

 mongosh	08-06-2024 19:04	Application	1,12,839 KB
 mongosh_crypt_v1.dll	08-06-2024 19:04	Application extens...	24,132 KB

```
C:\Users\ADMIN\Downloads\ x + v
Please enter a MongoDB connection string (Default: mongodb://localhost/): |
```

Step 5: Now Enter the mongosh on window.

```
mongosh mongodb://127.0.0.1:27017/ x + v
Please enter a MongoDB connection string (Default: mongodb://localhost/): mongosh
mongosh
Current Mongosh Log ID: 66840268ba0ef66a47cdcdf5
Connecting to:      mongodb://127.0.0.1:27017/mongosh?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.2.6
Using MongoDB:      7.0.9
Using Mongosh:      2.2.6

For mongosh info see: https://docs.mongodb.com/mongosh-shell/

-----
The server generated these startup warnings when booting
2024-07-02T18:56:46.320+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

mongosh> |
```

Few Commands

Command	Expected Output	Notes
show dbs	admin 40.00 KiB config 72.00 KiB db 128.00 KiB local 40.00 KiB	All Databases are shown
use db	switched to db db	Connect and use db
show collections	Students	Show all tables

To list the databases available to the user, use the helper show dbs.

```
mongosh> show dbs
admin      40.00 KiB
config     72.00 KiB
db         104.00 KiB
local     120.00 KiB
```

Switch Databases

To display the database you are using, type `use db`, to display collections type `show collections`.

```
mongosh> use db
switched to db db
db> show collections
candid
locations
products
student
student1
students
students_permission
db> |
```